

University of Sri Jayewardenepura

Department of Mathematics

MAT 2011

Linear Algebra II

Semester I 2026

Exercise Sheet 2

Note: Discussion of these exercises will be held on August 22nd, 2025. So, it is advisable to try these questions and join that session.

1. In problem I-VII determine whether the given matrix A is diagonalizable. If it is, find a matrix C such that $C^{-1}AC = D$.

I. $\begin{pmatrix} -2 & -2 \\ -5 & 1 \end{pmatrix}$

II. $\begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix}$

III. $\begin{pmatrix} 3 & 2 \\ -5 & 1 \end{pmatrix}$

IV. $\begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$

V. $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

VI. $\begin{pmatrix} 7 & -2 & -4 \\ 3 & 0 & -2 \\ 6 & -2 & -3 \end{pmatrix}$

VII. $\begin{pmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix}$

2. Show that if A is similar to B and B is similar to C , then A is similar to C .
3. If A is similar to B , show that A^n is similar to B^n for any positive integer n .
4. Let $D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ compute D^{20} .
5. If A is similar to B , show that $\det A = \det B$.
6. Suppose that $C^{-1}AC = D$. Show that for any integer n , $A^n = CD^nC^{-1}$. This gives an easy way to compute powers of a diagonalizable matrix.
7. Let $A = \begin{pmatrix} 3 & -4 \\ 2 & -3 \end{pmatrix}$. Compute A^{20} . [Hint: Find a C such that $A = CDC^{-1}$]

8. Let A be an $n \times n$ matrix whose characteristics equation is $(\lambda - c)^n = 0$. Show that A is diagonalizable if and only if $A = cI$.
9. Use the result of problem 6 to compute A^{10} , where $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$.
10. Let A and B be real $n \times n$ matrices with distinct eigenvalues. Prove that $AB = BA$ if and only if A and B have the same eigenvectors.
11. If A is diagonalizable, show that $\det A = \lambda_1 \lambda_2 \dots \lambda_n$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A .
12. Use Cayley_ Hamilton theorem to find the inverse of the 3x3 matrix $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$.



Reference: Elementary Linear Algebra by Grossman.